Boosting Medical Visual Understanding From Multi-Granular Language Learning

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Introduction

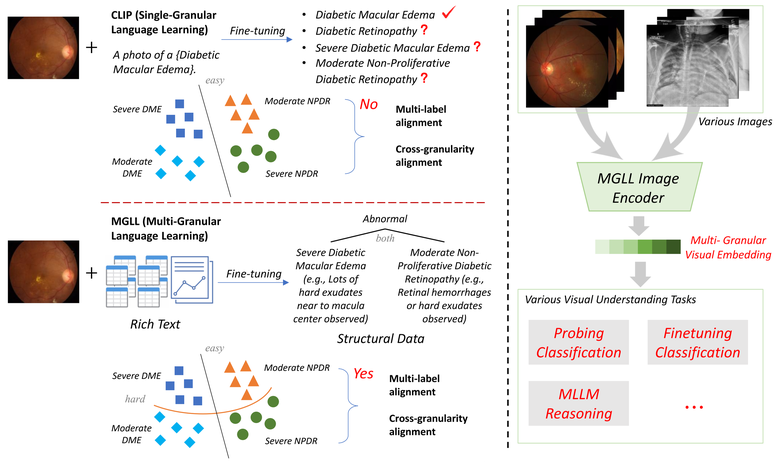

Figure 1. The illustrative comparison of data input and outcome between CLIP (Contrastive Language-Image Pre-Training) and MGLL (Multi-Granular Language Learning). DME (Diabetic Macular Edema), NPDR (Non-Proliferative Diabetic Retinopathy).

Most studies focus on single-label, single-granularity alignment limits its effectiveness in complex domains such as medical imaging, where images often correspond to multiple high-level labels (e.g., disease categories) across different annotation granularities (e.g., clinical explanation). To address this, we propose Multi-Granular Language Learning (MGLL) to improve both **multi-label** and **cross-granularity** alignment.

- We propose MGLL, a novel contrastive learning framework using multi-granular language that enables simultaneous multi-label and cross-granularity alignment.
- We provide a set of architecture-agnostic, multi-label, multi-granularity learning objectives that can be seamlessly integrated into foundation models to enhance visual understanding.
- We design a structured multi-granular, multi-label system and construct large-scale multi-granular retinal and X-ray 2D image-text datasets.

Method

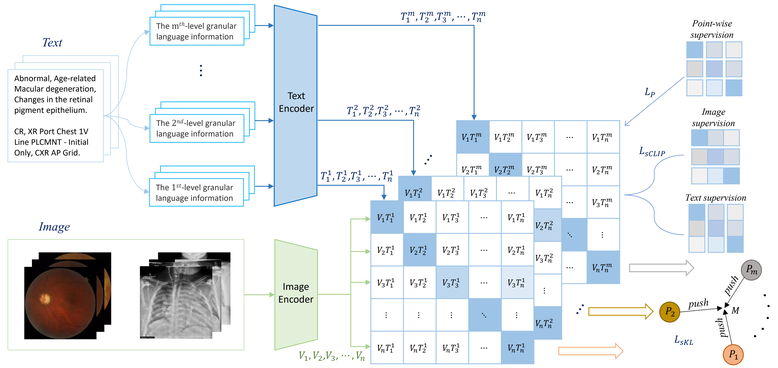

Figure 2. The overview of MGLL (Multi-Granular Language Learning), which includes the pretraining pipeline.

$$l_{ik} = -w_{ik} \log \frac{\exp(\text{sim}(V_i, T_{ik})/\tau)}{\sum_{n=1}^{N}\sum_{m=1}^{M_n} \exp(\text{sim}(V_i, T_{nm})/\tau)} \quad (1)$$

$$\mathcal{L}_{\text{sCLIP}} = \frac{1}{2\sum_{i=1}^{N} M_i} \sum_{i=1}^{N}\sum_{k=1}^{M_i} (l_{ik} + l_{ki}) \quad (2)$$

$$w_{ik} = \frac{\text{coocurrence}(V_i, T_{ik})}{\sum_k \text{coocurrence}(V_i, T_{ik})} \quad (3)$$

$$\mathcal{L}_{\text{P}} = -\sum_{i=1}^{N}\sum_{j=1}^{M} \frac{y_{ij}\log x'_{ij} + (1-y_{ij})\log(1-x'_{ij})}{N} \quad (4)$$

$$D_{\text{KL}}(P_i \| M) = \sum_j P_i^{(j)} \log \frac{P_i^{(j)}}{M^{(j)}} \quad (5)$$

$$\mathcal{L}_{\text{sKL}} = \sum_{i=1}^{m} D_{\text{KL}}(P_i \| M) \quad (6)$$

$$\mathcal{L}_{\text{MGLL}} = \alpha_1 \mathcal{L}_{\text{sCLIP}} + \alpha_2 \mathcal{L}_{\text{p}} + \alpha_3 \mathcal{L}_{\text{sKL}} \quad (7)$$

- **Soft CLIP loss** improves global image–text alignment and better supports multi-label representation learning.
- **Point-wise loss** strengthens alignment between visual features and text features at each specific granularity.
- **Smooth KL divergence loss** encourages features from different granularities into a shared feature space, improving cross-granularity consistency.
- We build large-scale multi-granularity image–text datasets (MGLL-Fundus and MGLL-Xray) for training and evaluation.
- **MGLL-Fundus** contains **246,389 image–text pairs** collected from **49 public datasets**, which covers **50+ retinal disease** categories.
- **MGLL-Xray** contains **190,882 image–text pairs** collected from the MIDRC repository, with multi-granularity textual annotations to support robust representation learning.

Table 1. The multi-granularity textual description examples of MGLL-Fundus dataset.

Disease Category	Clinical Explanation
Abnormal, Mild Non-Proliferative Diabetic Retinopathy Abnormal, Proliferative Diabetic Retinopathy Abnormal, Cataract Normal, Healthy	Only microaneurysms observed Neovascularization or vitreous/preretinal hemorrhage Opacification of crystalline lens observed Clear optic disk boundaries, Normal fundus color, No apparent retinopathy

Results

Table 2. Results on MIDRC-XR, MIDRC-XR-Portable, and ChestX-ray14. **Bold** indicates best and underline shows second-best.

Method	MIDRC-XR						MIDRC-XR-Portable						ChestX-ray14					
	Linear Probe (%)			Fully Fine-tune (%)			Linear Probe (%)			Fully Fine-tune (%)			Linear Probe (%)			Fully Fine-tune (%)		
	AUC	ACC	mAP	AUC	ACC	mAP	AUC	ACC	mAP	AUC	ACC	mAP	AUC	ACC	mAP	AUC	ACC	mAP
CLIP (ICML-21)	54.72	51.18	16.62	88.52	80.83	62.04	71.43	78.22	22.31	91.83	90.08	83.94	69.75	78.09	18.33	82.05	87.58	31.79
CheXzero (Nat. BME-22)	51.31	43.85	12.71	80.11	73.26	55.46	72.84	80.13	23.56	92.47	92.42	85.23	68.72	76.98	15.97	81.81	87.39	31.52
KAD (Nat. Com-23)	53.44	47.13	14.86	85.74	78.39	60.12	73.53	80.71	23.88	93.41	92.98	85.96	73.72	78.95	21.87	83.80	89.13	34.01
MRM (ICLR-23)	56.23	53.61	17.73	90.67	83.95	64.76	79.38	86.05	27.72	96.52	95.07	86.95	74.63	79.87	23.23	84.28	89.57	35.62
UniChest (TMI-24)	59.02	54.78	19.32	92.51	86.32	66.93	78.49	85.28	27.37	95.44	94.32	86.38	76.15	81.72	25.52	85.84	89.99	37.97
UniMed-CLIP (arXiv-24)	57.33	54.07	18.06	94.15	87.47	68.49	80.05	86.63	28.16	94.31	93.55	86.19	75.54	81.21	24.96	82.59	88.36	32.19
CARZero (CVPR-24)	57.92	54.43	18.79	93.48	86.94	67.62	75.24	82.67	25.65	92.94	92.66	85.67	77.32	83.94	26.88	82.95	88.65	32.86
FG-CLIP (ICML-25)	58.31	54.59	19.03	93.29	86.71	67.44	80.31	86.77	28.27	96.93	95.74	87.42	76.62	82.35	25.98	85.10	89.73	37.02
MGLL	**61.25**	**56.57**	**21.19**	**99.08**	**90.06**	**73.33**	**83.86**	**89.06**	**30.62**	**99.75**	**98.80**	**89.87**	**82.94**	**90.41**	**28.53**	**87.37**	**92.71**	**39.17**

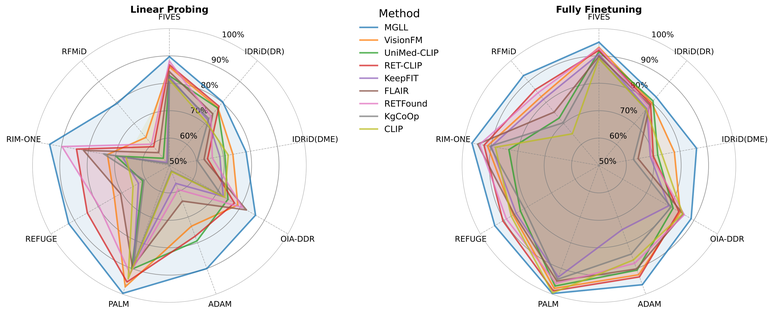

Figure 3. The quantitative comparison (AUC) between baseline methods and proposed MGLL on nine downstream datasets.

- On **nine downstream retinal datasets** (Figure 3), MGLL achieves the best results in both **linear probing** and **full fine-tuning**. The largest improvement is observed on the **multi-label RFMiD dataset**, where MGLL exceeds other methods by at least **16.6% AUC** in linear probing and **6.7% AUC** in full fine-tuning.
- On MIDRC-XR and MIDRC-XR-Portable, MGLL improves AUC by **2.23%** and **3.81%** over the second-best methods in linear probing, with an even larger advantage after full fine-tuning.

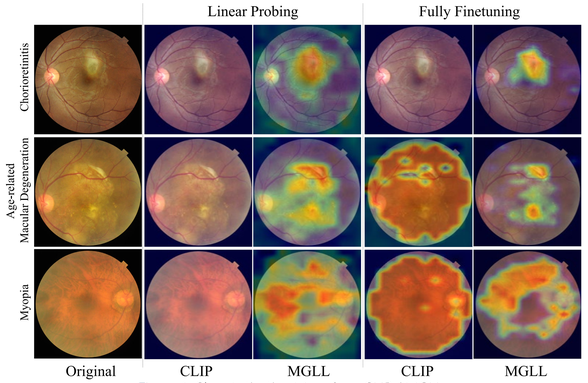

Figure 4. Class Activation Maps from CLIP / MGLL.

- Qualitative CAM (Figure 4) show that MGLL localizes disease-relevant regions more accurately.
- Specifically, MGLL highlights hard exudates for chorioretinitis, the retinal pigment epithelium for age-related macular degeneration, and leopard fundus regions for myopia.

Table 3. Comparison of multiple-choice accuracy with MGLL in multimodal large language models on selected ten diseases.

Method	AMD	Cataract	CSR	DR	Glaucoma	Media Haze	Myopia	Retinitis	DME	Tessellation	Average ↑
InstructBLIP	80.17%	80.00%	0.00%	76.51%	59.30%	16.13%	44.25%	11.11%	63.79%	41.67%	47.29%
+ MGLL	**83.63%**	**85.00%**	**28.57%**	**82.55%**	**65.43%**	**45.16%**	**52.65%**	**44.44%**	**74.14%**	**58.33%**	**61.99%** (14.7% ↑)
Mini-Gemini	76.61%	85.00%	14.29%	79.87%	67.90%	38.71%	58.41%	33.33%	60.34%	33.33%	54.78%
+ MGLL	**82.46%**	**85.00%**	**42.86%**	**84.56%**	**72.22%**	**58.06%**	**64.16%**	**55.56%**	**65.52%**	**41.67%**	**65.21%** (10.4% ↑)
Qwen-VL	81.87%	75.00%	28.57%	80.54%	78.40%	54.84%	76.55%	22.22%	84.48%	25.00%	60.75%
+ MGLL	**85.96%**	**80.00%**	**42.86%**	**89.93%**	**87.04%**	**70.97%**	**80.97%**	**33.33%**	**89.66%**	**41.67%**	**70.24%** (9.5% ↑)
InternVL	81.29%	85.00%	71.43%	94.63%	89.51%	64.52%	88.05%	44.44%	87.93%	66.67%	77.35%
+ MGLL	**86.55%**	**90.00%**	**71.43%**	**96.64%**	**90.74%**	**67.74%**	**91.15%**	**55.56%**	**94.83%**	**75.00%**	**81.96%** (4.6% ↑)
LLaVA	83.04%	90.00%	42.86%	87.25%	91.36%	48.39%	88.50%	44.44%	93.10%	58.33%	72.73%
+ MGLL	**84.80%**	**90.00%**	**57.14%**	**93.96%**	**91.98%**	**61.29%**	**90.71%**	**66.67%**	**96.55%**	**66.67%**	**79.98%** (7.3% ↑)
LLaVA-Med	16.37%	15.00%	42.86%	26.85%	25.31%	25.81%	23.89%	33.33%	16.67%	16.67%	24.28%
+ MGLL	**58.48%**	**65.00%**	**57.14%**	**77.18%**	**59.26%**	**51.61%**	**57.08%**	**44.44%**	**55.17%**	**58.33%**	**58.37%** (34.1% ↑)
Med-Flamingo	25.73%	30.00%	57.14%	36.91%	24.07%	22.58%	18.58%	22.22%	24.14%	8.33%	26.97%
+ MGLL	**69.01%**	**75.00%**	**71.43%**	**80.54%**	**61.11%**	**54.84%**	**45.58%**	**44.44%**	**51.72%**	**33.33%**	**58.70%** (31.7% ↑)
Janus-Pro	88.30%	75.00%	42.86%	93.29%	90.74%	58.06%	87.17%	33.33%	62.07%	58.33%	68.92%
+ MGLL	**90.64%**	**85.00%**	**71.43%**	**96.64%**	**95.06%**	**67.74%**	**90.27%**	**55.56%**	**70.69%**	**75.00%**	**79.80%** (10.88% ↑)

- MGLL brings **consistent and substantial performance gains** across all tested MLLMs.
- The average accuracy improvement ranges from **4.6%** for **InternVL** to **34.1%** for **LLaVA-Med**.

Conclusion

We introduce Multi-Granular Language Learning (MGLL), a contrastive learning framework that leverages textual information at multiple granularities to improve vision-language representations. MGLL leverages textual information at multiple granular levels to enhance representation quality and captures complex semantics effectively for 2D medical images, with potential extensions to 3D imaging such as CT and MRI.

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